

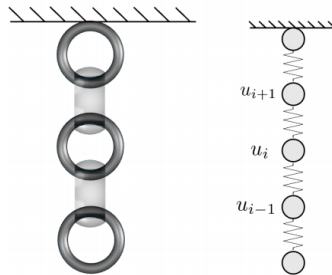
Background info and Motivation

Polycatenated architected materials (PAMs) are a new class of metamaterials composed of interlocked, ring-like units whose mechanical behavior is governed by both constituent material properties as well as the unit/linkage topology. The high internal degrees of freedom from these interlocking connections give rise to nonlinear force-displacement relationships, enabling PAMs to exhibit tunable stiffness, energy dissipation, and wave propagation characteristics. Prior studies have primarily focused on the static and quasi-static mechanical response of PAMs. However, understanding their dynamic behavior (especially how mechanical impulses propagate through chains and networks of PAM units) is essential for use in applications such as shock absorption, impact mitigation, and mechanical signal control.

Theoretical models model these as masses on nonlinear springs (at the linkages), which predict that under appropriate conditions, PAM chains can support the propagation of solitons, which are localized, particle-like excitations. My previously work aimed to observe and characterize these dynamics experimentally to provide validation of these models, however, it ended up as a study on the more wave-like behavior of PAMs and was unable to consistently generate particle-like behavior. The work done this quarter builds on the previously developed experimental platform to investigate the conditions required for soliton formation and to eventually extend these to more complex geometries, such as chain junctions.

New Work and Findings

Solitons are predicted from strongly nonlinear systems, such as ones described by the KdV equation which can model waves in granular particles, which behave very similarly to PAMs. Upon discussing with my mentor, we found a potential reason why we did not see the particle-behavior based on the theoretical model.



$$F_{net} = m\ddot{u}_i = k(u_{i+1} - u_i + \delta_T)^n - k(u_i - u_{i-1} + \delta_T)^n$$

The above equation describes the motion of one unit in the nonlinear springs model, where u_i is the displacement of the i -th mass from its equilibrium position, $\ddot{u}_i$ is acceleration of the i -th mass, and δ_T is the displacement due to pretension. By making a long wavelength approximation and, crucially, by taking the limit as the displacement due to the excitation

amplitude ($u_i - u_{i-1}$) is much greater than the displacement due to pretension, the equation takes on a form similar to the KdV equation: yielding soliton solutions.

In other words, high excitation amplitude with low pretension on the system is required to observe soliton behavior. The current set up for exciting the PAM sample was apparently unable to apply an input amplitude high enough for this approximation to hold. As such, the quarter was focused on making a new actuation method with the hopes of seeing soliton particle like behavior instead of wavelike behavior.

After we succeed in actuating a soliton into the PAM sample, we hope to investigate its properties in a junction set up where 1D PAM chains split into two at an angle, to investigate its behavior when PAMs are linked at an angle. This is relevant for investigating how forces/impacts distribute in 2D or 3D PAMs both for practical applications as well as a study of complex nonlinear systems.

We were able to (very roughly) see this soliton behavior by manually exciting the sample, hitting one of the ends holding the PAM chain with the back of a screwdriver. The momentum transferred by the swinging striker and the short interaction time in the collision was enough to create a sharp, short impulse as desired. The issue with this is that a manual striker is far too inconsistent and unreliable to produce repeatable experimental data.

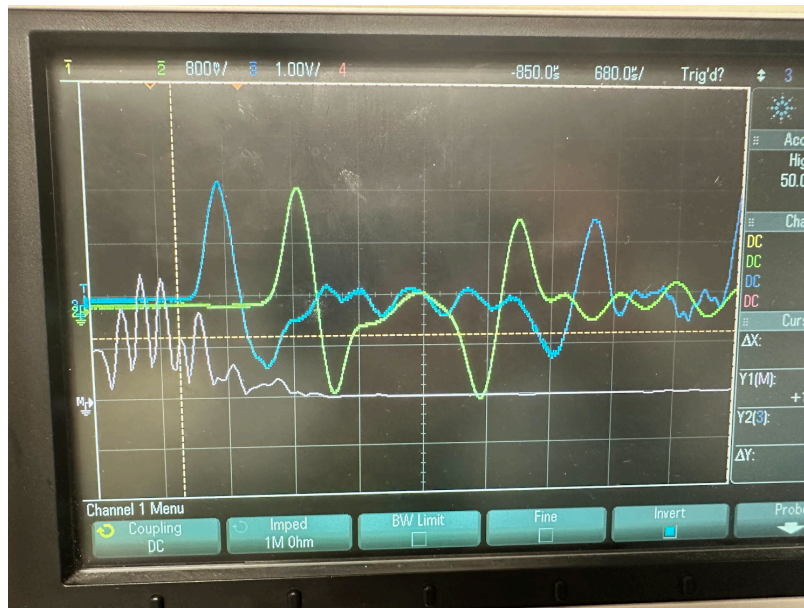


Fig 1. Accelerometer data from 2 PAM units along a chain showing an example of particle like behavior: one big pulse with very little waves following it (after the middle of the screen, the pulses that got reflected from the fixed boundary reach the sensors again). This was done with manual actuation using the back of a screwdriver.

To do this more reliably, we modified the setup to allow for lower pretensions and created and tested two ideas for a mechanical mechanism for high amplitude short time duration

actuation. Previously, very low pretensions below 3N were not supported with the setup because it caused too much dangling of the PAM sample due to its own mass. Dangling is problematic because it has a non-negligible angle between the contacts of multiple units, making the theory-based simulation the lab had inaccurate in predicting pulse speeds.

We solved this by simply adding hooks into the design for each PAM unit, and tensioning a thin fishing string across the PAM sample. This way, the chain would no longer dangle, and we would have a straightly aligned and appropriately tensioned PAM chain even for smaller pretensions.

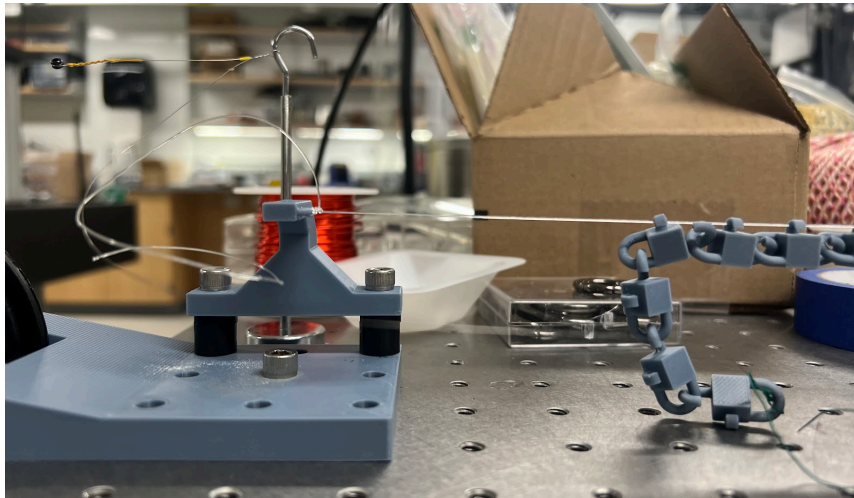


Fig 2. The string used to hold the PAM chain on the right straight to avoid draping.

The more difficult challenge was the different actuation method, especially since the piezo electric actuator was already on the millisecond level, but not enough for a relatively “high” amplitude actuation. We took inspiration from the experimental methods for granular systems, where, for a line of spherical units contacting each other, a striking sphere of the same mass is dropped from a ramp to strike the target at one end. This successfully applies a short but high amplitude pulse that travels through the spherical samples, allowing experimentalists to investigate the properties of granular materials. In our case, however, we wish to investigate the more unique properties of PAMs: solitons in the compression pulse (pushing) direction are already observed and studied in granular particles, but tension (pulling) pulses are only possible for PAMs.

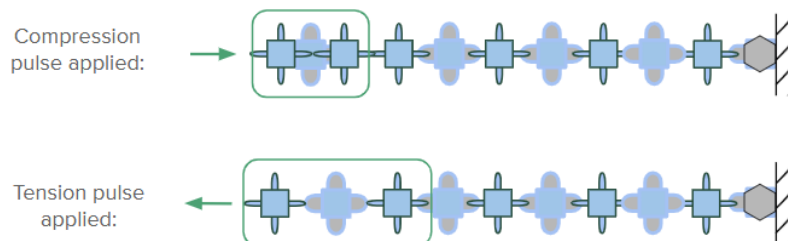


Fig 3. An image showing the difference between a compression and tension pulse.

As such, we needed to modify the method for granular particles to be able to excite tension pulses instead, while still being able to control what amplitude impulse we can apply. One way we attempted this is to attach one end of the chain to a string going through a pulley connected to a mass system pulled down by gravity. This mass system consists of 2 separate masses: a bottom mass constantly applying pretension on the string, and another elevated to a certain height above the bottom one, such that when it is dropped from the height, the top mass collides with the bottom mass and ideally applies a sharp impulse to the string. We created a rough version of this to test as imaged in figure 4.

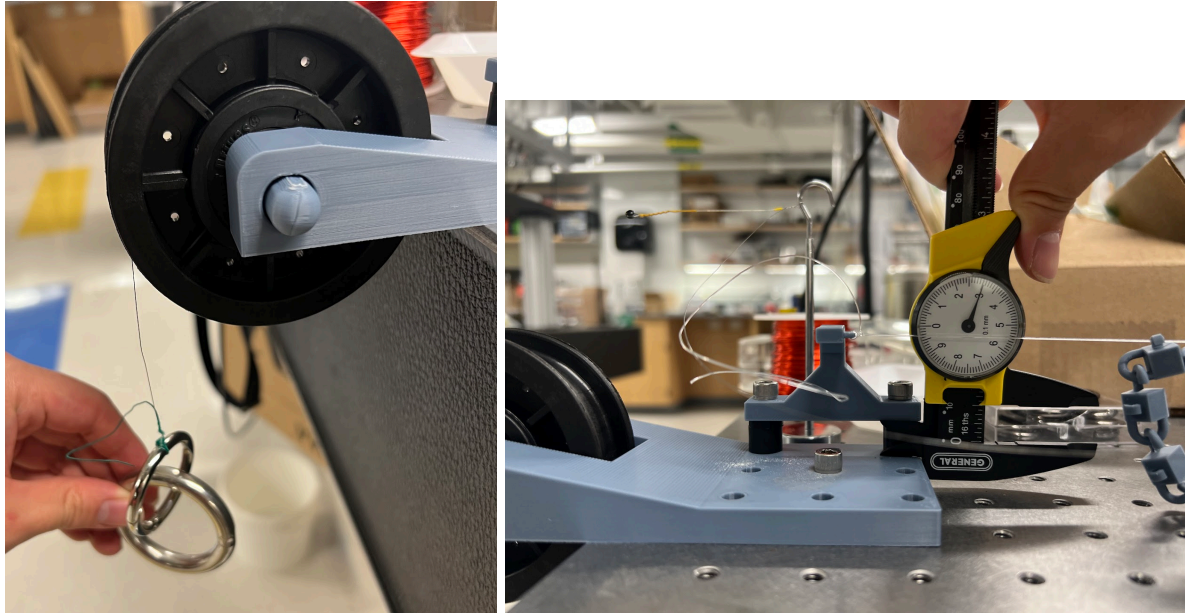


Fig 4. Images of the pulley mechanism being tested and measured. In the image, a rough test is done with 2 rings, but another test was performed with 2 cylindrical masses on a single vertical rod.

This was unfortunately unsuccessful. The top mass's collision with the bottom mass caused too much ringing and inconsistencies. More significantly, the elasticity of the string used between the chain and the masses also added some slack, which transferred the impulse more slowly (possibly with some damping) and decreased the impulse's sharpness. We tested this again with a metal wire instead, but the same issue persisted. The resulting impulse sent into the system looked more like waves instead of a single sharp pulse with no ringing following it. We realized from this that the striker should strike the end of the PAM chain as directly as possible, without any intermediate materials.

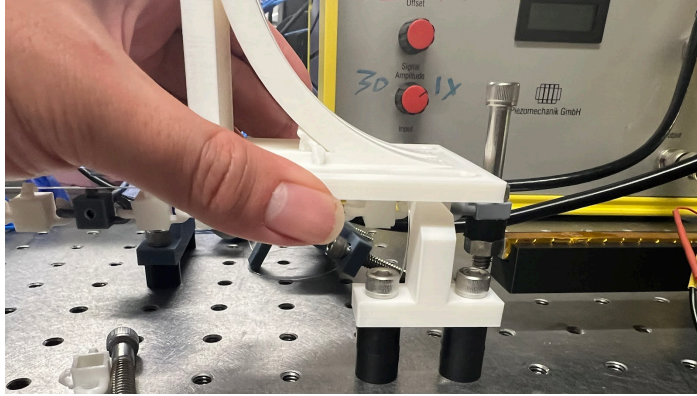


Fig 5. Image of prototype for a second design where a ball striker goes down the ramp to hit an extended screw attached to the end of the PAM chain, going under the ramp and hand and towards the left in the image.

In our second design, we essentially used the same mechanism as the granular particle striker design, but pointed in the opposite direction. The end of the PAM chain is held by a longer screw, which is hit from the inside of the PAM chain to the outward direction by a ball striker descending a ramp. The placement of the striker on the ramp would determine the input amplitude. With this method, we saw some potential for success although it was limited. Different striker materials had different properties: metal ones would supply higher pulse amplitudes and sharper peaks but have high frequency ringing, possibly due to the metal-metal collision. Plastic ball strikers yielded smoother peaks but were much smaller amplitude due to its lower weight.

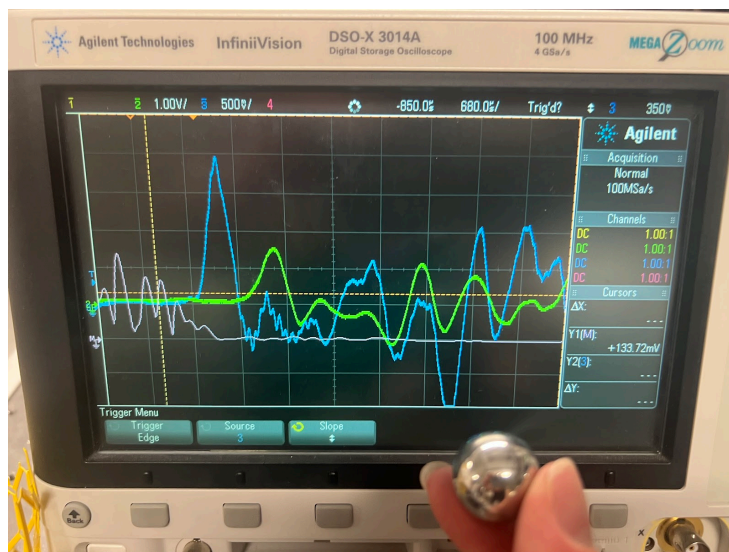


Fig 6. Result of a metal ball striker: high, single pulse as desired but with high frequency ringing as an artifact. Later pulses are due to reflected pulses from the fixed boundary

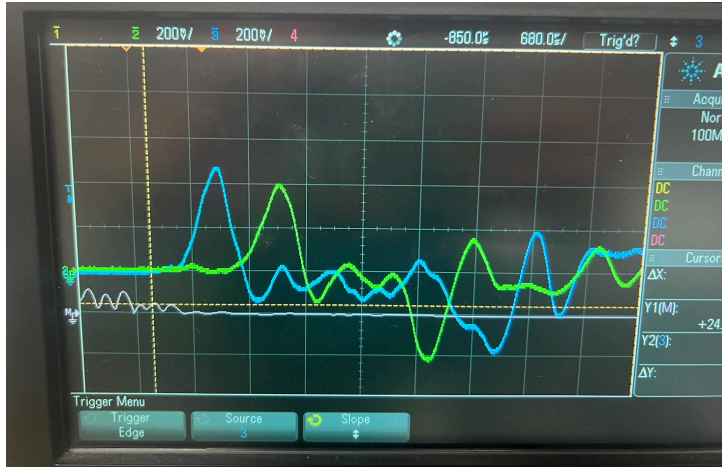


Fig 7. Result of a plastic ball striker: much smoother peaks with somewhat particle-like behavior, but with a much lower amplitude and relatively higher “wave-like” behavior.

Based on the rule of thumb for granular particle system excitation and momentum conservation, a striker should ideally have a similar mass to each unit of the system for optimal momentum transfer from the striker to the unit. Achieving this with this system may be difficult due to the light weight of each unit ($\sim 5\text{g}$), and how a very light striker ball's max velocity would be limited by friction and air resistance.

As next steps, we plan to explore this option by increasing ramp size and testing other striker masses, but in parallel, develop another design: a rotating hammer-like striker actuated by elastic bands to achieve higher velocities for lower masses. This is also similar to the rough “manual” excitation method we initially used.